

## Guide to Writing Lab Reports

Your lab report is commonly known as a technical report, a concise way of communicating what you found from a technical investigation. It is the most common type of communication within industry, business, laboratories, and researcher. It is the way to disseminate information to your colleagues both in the present and the future. In other words, your audience is people who have the same general background & concerns that you do but do not have any detailed knowledge of what you did and what you found out. This is an informational document that is much different than a publication in an academic or trade journal. It is as brief as possible to communicate the important details including motivations and results. To show that you value your colleagues, the report should not include superfluous language, weasel words, or off topic meandering. The format of every report should be “Tell them what you will tell them”; “Tell them”; “Tell them what you told them”.

### Introduction

Every report begins with a short introduction. Its purpose is to tell a reader why they should read the rest of the report. Your introduction should just be a few short paragraphs. It should give the motivation for your study, how it connects to previous knowledge, its conclusion, and what will be addressed in the report. **This is the most important part of any technical report.**

You should include the following three things in your lab report introduction:

1. What question are you trying to answer.
2. Why you are trying to answer that question.
3. What physics did you use to answer the question.

### Prediction

In physics, your prediction is typically based on fundamental principles often expressed as equations and graphs. This section should include your knowledge of the physics theory that motivated your measurements. It should reflect the knowledge you have gained from the components of the course. You **must** justify your prediction with physics concepts. This section tells the reader about your physics knowledge before you completed the lab. Your results do not have to agree with your prediction, but your conclusion must include any revision to your physics knowledge based on your lab results.

Your prediction section should include:

1. A proposed answer to the question of posed by the lab (the prediction), usually in the form of an equation and a graph.
2. A detailed justification of your prediction with fundamental physics.

## Procedure

This section **is the second most important part of your lab report**. It should have enough detail for someone to reproduce what you did but not include obvious or unnecessary steps.

You should include the following in your procedure section:

1. Enough detail to allow someone with your background and knowledge to replicate your results.
2. A description of how you determined the uncertainty of your measurements.
3. An identification of any systematic errors in your measurements that may have affected your result.

## Analysis

In this section, build your case as to why you accept or reject the prediction you made earlier in the report. It should include the relevant measurements presented in tables and graphs with uncertainties. You should describe any calculations you did using your measurements and why they are appropriate. Be quantitative and definite. Avoid weasel words such as “very close to”, “about”, “larger than” or “human error”. **Because the analysis should be completed before you leave the lab, there should be no missing or incorrect data.** If something is wrong with the data, you must explain why you did not catch and correct it in the lab. If you don’t believe your measurements but don’t know what went wrong, state the reason you don’t believe the results. **Split your analysis discussion into distinctive sections such as in the examples below.** Each section should be as brief as possible. Strive for no more than 3 paragraphs per section.

### Section 1: Results and Uncertainties

Present your measurements, including their uncertainty, and discuss how those measurements give your result. Use tables and graphs. Discuss the important numerical uncertainties of your measurements and how that uncertainty affects your interpretation of the result. Below are a couple of examples.

Example 1: “Since the uncertainty for the acceleration measurements was  $\pm 0.2 \text{ m/s}^2$ , the measured accelerations of the tennis ball ( $9.7 \text{ m/s}^2$ ) and the lacrosse ball ( $9.8 \text{ m/s}^2$ ) are not significantly different. Therefore, our data is consistent with the hypothesis that the acceleration of a dropped ball does not depend on its mass.”

Example 2: “As shown in Table 2, the speed of cart A was  $5.2 \pm 0.1 \text{ m/s}$ , and the speed of cart B was  $2.1 \pm 0.1 \text{ m/s}$ . Based on the uncertainty of our measurements, the minimum speed of cart A could be  $5.1 \text{ m/s}$  and the maximum speed of cart B could be  $2.2 \text{ m/s}$ . Therefore, our data shows the speed of cart A is significantly greater than the speed of cart B.”

## Section 2: Uncertainty

Discuss the sources of the relevant measurement uncertainties and estimate possible sources of systematic error caused by your equipment or experimental technique. Discuss how you arrived at your systematic error estimate and your measurement uncertainties. Also discuss how these uncertainties and errors could affect your conclusion. Suggest specific changes that could improve the measurements used test of your prediction.

Example 1 (systematic error): “One potential source of error is that the initial calibration of the video could have been incorrect because the placement of the meter stick was not always near the actual path of the ball. The average acceleration of the ball was measured to be  $10.9 \pm 0.4 \text{ m/s}^2$ , which does not agree with the known gravitational acceleration of  $9.8 \pm 0.1 \text{ m/s}^2$  as given by Wikipedia. If the meter stick image in the video was placed so that distances marked using the video were 10% smaller than the actual changes, all the measured accelerations would be increased by this factor. An improvement in experimental technique would be to analyze the motion of a dense ball to get its acceleration and compare it to the known gravitational acceleration before measuring any other balls. If there were a significant difference of that acceleration with that expected, tests should be run to determine the effect of changing the position of the meter stick and position of the camera.”

Example 2 (random error): “A potential source of error could have been inconsistently tracking the cart in the video by not marking the image on the video frames at the same spot. This results in the recorded distance traveled for each time interval to shift randomly causing an uncertainty in the calculated acceleration. An improvement in experimental technique would be to put a spot on a piece of tape on the cart and always record the position.”

## Section 3: The Answer

In the third section, **answer the question** posed in the introduction based on your measurements and calculations and **discuss the reasonableness of that**. Connect these physics ideas used to the fundamental physics being discussed in the course. Additionally, if your prediction was does not agree with your result, discuss potential factors that were not considered in the prediction. Be specific and quantitative. For example, if you ignored air resistance in your prediction but think it could make a difference, use your data or other sources of information to estimate how much it could have impacted your results.

## Graphs and Tables

Throughout this section, you should include relevant, clearly labeled graphs and figures. Only include what is helpful and relevant to the reader. Remember to include **units and uncertainties for all measurements** and **include error bars on all plots**. Make sure to include an identification (i.e. “Figure 1”) and descriptive caption for all plots and tables and use this identification to refer to the figures in the text of your report. Label the axes on all graphs that you include.

## **Conclusion**

This section should be one or two paragraphs wrapping up your report. It is a brief recap of information that is already in your report. Compare your prediction to your result. State and justify whether your measurements support your prediction, refute your prediction, or are inconclusive. State what you would do to improve your measurements if you needed to repeat the experiment.

You should include the following (likely in this order) in your conclusion:

1. A summary of what you did and the purpose of this report.
2. Your result.
3. A comparison of your result to your prediction and the implications of that comparison.
4. A summary of your suggested specific improvements to the equipment, the experimental technique, or the analysis technique used.

## **General Formatting Notes**

- Writing a concise and complete technical report comes with practice. Your report should be no more than 4 pages. A shorter but still complete report is even better. Don't worry if yours is too long at the beginning. Improvement comes with practice.
- Your report should be spaced to be readable with enough space for your TA to make constructive comments.
- Graphics should be clear and easy to read with appropriate titles, labels, error bars, and descriptive captions.
- If you include an equation in your report, put it on its own line and make sure to define the symbols in the equation.
- Keep a consistent grammatical tense throughout the report.
- Use professional language.
- Keep writing concise.
- Use a spell checker and grammar checker.
- Read your report all the way through before turning it in! This helps catch minor errors and left out phrases to make sure the report flows well. It is even better if you can get someone else to read it and give you feedback.
- Remember the structure of every paper: "Tell them what you are going to tell them. Tell them. Tell them what you told them."